

07AMO5

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Problem

Prove that for every nonnegative integer n , the number $7^{7^n} + 1$ is the product of at least $2n + 3$ (not necessarily distinct) primes.

Solution. We will perform induction on n .

For $n = 1$, $7^7 + 1 \equiv 0 \pmod{4}$ and since $7^7 + 1 > 4$, there must be a third prime dividing it, hence making it have at least 3 prime divisors.

Assume that this works for $n = k$, we will prove it for $n = k + 1$.

Since $7^{7^{k+1}} + 1 = 7^{7 \cdot 7^k} + 1 = (7^{7^k} + 1)(1 - 7^{7^k} + 7^{2 \cdot 7^k} - \dots + 7^{6 \cdot 7^k})$, it is enough to show that $1 - 7^{7^k} + 7^{2 \cdot 7^k} - \dots + 7^{6 \cdot 7^k}$ has at least two prime factors.

We have: $X^6 - X^5 + X^4 - \dots - X + 1 = (X + 1)^6 - 7X(X^2 + X + 1)^2$. For $X = 7^{7^k}$, it is the difference of two squares. Since difference of two squares has two factors at least, we know that $1 - 7^{7^k} + 7^{2 \cdot 7^k} - \dots + 7^{6 \cdot 7^k}$ must have 2 prime factors at least.

We have $2n + 3 + 2 = 2(n + 1) + 3$. This finishes the inductive step.